Question	Answer	Marks	Partial Marks
7(c)	64	**1**	**FT** *their* **(b)** ÷ 2
7(d)	116	**1**	**FT** 180 – *their* **(c)**
8(a)	370 or 370.2 to 370.3	**2**	**M1** for 864 ÷ *their* time
8(b)	991 or 990.5 …	**4**	**M2** for $864^2 + 928^2 - 2 \times 864 \times 928\cos 67$ or **M1** for correct implicit version **A1** for 981100 to 981110
8(c)(i)	313	**2**	**M1** for 180 + 133 or 360 – 47
8(c)(ii)	[0]79.5 to [0]79.6 …	**4**	**M2** for $\frac{928 \times \sin 67}{\textit{their } 991}$ or $\frac{864 \times \sin 67}{\textit{their } 991}$ oe or **M1** for implicit form of either **A1** for [angle *HGB* =] 59.5 to 59.6 … or [angle *HBG* =] 53.4 or 53.37 to 53.42 **M1 dep** for *their* angle *HGB* + 20 leading to answer or for 133 – *their* angle *HBG* leading to answer
9(a)(i)	42.8 or 42.79 … nfww	**4**	**M1** for mid-values soi **M1** for Σ*fm* where *m* is any value in interval including boundaries **M1** (dep on second **M1**) for *their* Σ*fm* ÷ 120
9(a)(ii)	Blocks of height 1.8 4.4 8 2.1 with correct widths	**4**	**B1** for each correct block If **B0**, **SC1** for correct frequency densities seen
9(b)	Valid general comment about distributions	**1**	e.g. [On average], shoppers spend less time shopping on Wednesday oe
10(a)(i)	75 000 × 60 × 20 oe	**M1**	Allow × 1200 for × 60 × 20
10(a)(ii)	16.4 or 16.36 …	**3**	**M2** for $\frac{9 \times 10^7 \times 100}{1000 \times 55 \times 10^4}$ oe or **B2** for answer 0.164 or 0.1636 … or **B1** for answer figs 164 or 1636 … or **M1** for figs 9 ÷ figs 55
10(a)(iii)	28.3 or 28.27 to 28.28	**3**	**M2** for $\frac{76}{360} \times 2\pi \times 8.5 + 2 \times 8.5$ oe or **M1** for $\frac{76}{360} \times 2\pi \times 8.5$ oe

Question	Answer	Marks	Partial Marks
10(b)(i)	3770 or 3769 to 3770. …	**2**	**M1** for $\frac{1}{3}\times\pi\times10^2\times36$
10(b)(ii)	3.68 or 3.683 to 3.684 …	**4**	**M3** for $[r^3=]\ \frac{1}{2}\times$ *their* **(b)(i)** $\times\ \frac{3}{4\pi\times9}$ oe or **M2** for $\frac{4\pi r^3}{3}+\frac{4\pi(2r)^3}{3}=\frac{1}{2}\times$ *their* **(b)(i)** or for $\frac{4\pi r^3}{3}=\frac{1}{1+8}\times\frac{1}{2}\times$ *their* **(b)(i)** or **M1** for $\frac{4\pi r^3}{3}+\frac{4\pi(2r)^3}{3}$ or $\frac{1}{2}\times\frac{\pi\times10^2\times36}{3}$ or $\frac{1}{2}$ *their* **(b)(i)** seen or ratio of vols = 1 : 2^3 oe seen
11(a)(i)	$\begin{pmatrix}-19\\-2\end{pmatrix}$	**2**	**B1** for answer $\begin{pmatrix}-19\\k\end{pmatrix}$ or $\begin{pmatrix}k\\-2\end{pmatrix}$ or for $\begin{pmatrix}-9\\6\end{pmatrix}$ or $\pm\begin{pmatrix}10\\8\end{pmatrix}$ seen
11(a)(ii)	3.61 or 3.605 to 3.606	**2**	**M1** for $\sqrt{([-]3)^2+2^2}$ oe
11(a)(iii)	$-3m+5n=14$ and $2m+4n=9$	**B1**	Accept equivalents
	$[m=]-\frac{1}{2}$ or -0.5 and $[n=]\ 2\frac{1}{2}$ or 2.5 or $\frac{5}{2}$ with evidence of a correct algebraic method	**4**	**M1** for correctly equating one set of coefficients of *their* equations or rearranges one of *their* equations to make *m* or *n* the subject e.g. $[m=]\ \frac{1}{2}(9-4n)$ oe **M1** for correct method to eliminate one variable for *their* equations or correctly substitutes *their m* or *their n* into the other equation e.g. $-\frac{3(9-4n)}{2}+5n=14$ oe **B1** for one correct answer
11(b)(i)(a)	$-\mathbf{a}+2\mathbf{c}$	**1**	
11(b)(i)(b)	$\frac{3}{8}(-\mathbf{a}+2\mathbf{c})$ or $-\frac{3}{8}\mathbf{a}+\frac{3}{4}\mathbf{c}$ oe	**1**	**FT** $\frac{3}{8}$ (*their* **(b)(i)(a)**) in simplest form